

Day 26 – Screening Scoring Engine

1. Overview

Day 26 focused on developing the **Screening Scoring Engine** for the Zecpath AI recruitment platform.

The objective was to transform candidate interview responses into measurable screening scores and generate an overall candidate performance assessment.

The scoring system evaluates each candidate response using four parameters:

- **Clarity**
- **Relevance**
- **Completeness**
- **Consistency**

Each parameter is assigned a maximum score of **25**, giving every screening question a maximum score of **100**.

The completed system consists of six implementation steps:

1. Scoring Parameter Definition
2. Per-Question Scoring
3. Score Normalization
4. Overall Screening Score
5. Score Interpretation
6. Final Screening Score Integration

2. Day 26 Objectives

The main objectives of Day 26 were:

- Define standardized screening evaluation parameters.
- Calculate scores for individual screening questions.
- Validate scoring ranges.
- Normalize scores to a percentage.
- Calculate the candidate's overall screening score.
- Interpret the final score into meaningful performance categories.

- Integrate all scoring components into a final screening evaluation engine.
 - Test the complete scoring workflow using strong, average, and weak candidate scenarios.
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3. Screening Scoring Parameters

The first step was to define the parameters used to evaluate candidate responses.

Each parameter has a maximum score of **25**.

Clarity

Measures how clearly the candidate communicates the answer.

Maximum Score: 25

Relevance

Measures how relevant the answer is to the screening question.

Maximum Score: 25

Completeness

Measures whether the candidate provides sufficient information.

Maximum Score: 25

Consistency

Measures whether the answer is consistent with the candidate's other responses.

Maximum Score: 25

Therefore:

Maximum score per question = 25 + 25 + 25 + 25 = 100

4. Step 1 – Scoring Parameter Configuration

The scoring parameters were implemented as a structured configuration:

SCORING_PARAMETERS = {

 "clarity": {

 "max_score": 25,

 "description": "Measures how clearly the candidate communicates the answer."

```
},
```

```
"relevance": {
```

```
    "max_score": 25,
```

```
    "description": "Measures how relevant the answer is to the screening question."
```

```
},
```

```
"completeness": {
```

```
    "max_score": 25,
```

```
    "description": "Measures whether the candidate provides sufficient information."
```

```
},
```

```
"consistency": {
```

```
    "max_score": 25,
```

```
    "description": "Measures whether the answer is consistent with the candidate's other  
responses."
```

```
}
```

```
}
```

A helper function was also implemented:

```
def get_scoring_parameters():
```

```
    return SCORING_PARAMETERS
```

This allows other modules to retrieve the configured scoring parameters.

Step 1 Validation

The parameter configuration was successfully tested.

DAY 26 - STEP 1 SCORING PARAMETERS TEST

Clarity:

Maximum Score: 25

Relevance:

Maximum Score: 25

Completeness:

Maximum Score: 25

Consistency:

Maximum Score: 25

STEP 1 TEST: PASSED

5. Step 2 – Per-Question Scoring

The second step implemented scoring for individual screening questions.

The function accepts four scores:

```
calculate_question_score(  
    clarity,  
    relevance,  
    completeness,  
    consistency  
)
```

Each score must:

- Be numeric.
- Be greater than or equal to 0.
- Be less than or equal to 25.

The total question score is calculated as:

Total Score =

Clarity +

Relevance +

Completeness +

Consistency

The maximum score is:

100

6. Per-Question Validation

Three candidate response scenarios were tested.

Strong Answer

Clarity: 23

Relevance: 24

Completeness: 22

Consistency: 24

Total Score: 93 / 100

Average Answer

Clarity: 18

Relevance: 17

Completeness: 16

Consistency: 18

Total Score: 69 / 100

Weak Answer

Clarity: 10

Relevance: 8

Completeness: 9

Consistency: 7

Total Score: 34 / 100

Invalid score validation was also tested successfully.

Test 4 - Invalid Score: PASSED

STEP 2 TEST: PASSED

7. Step 3 – Score Normalization

The third step implemented score normalization.

Normalization converts scores with different maximum values into a common **0–100 percentage scale**.

The formula used is:

Normalized Score =

$(\text{Obtained Score} / \text{Maximum Score}) \times 100$

Example 1

$93 / 100 \times 100 = 93.0$

Example 2

$45 / 50 \times 100 = 90.0$

Example 3

$75 / 80 \times 100 = 93.75$

Zero Score

$0 / 100 \times 100 = 0.0$

Invalid score and invalid maximum-score conditions were also tested.

Test 5 - Invalid Score: PASSED

Test 6 - Invalid Maximum Score: PASSED

STEP 3 TEST: PASSED

8. Step 4 – Overall Screening Score

The fourth step calculates the candidate's overall screening score across multiple questions.

For each question, the four scoring parameters produce a maximum of **100 points**.

For three questions:

Maximum Score = $3 \times 100 = 300$

The overall score is calculated using:

Overall Score =

$(\text{Total Score} / \text{Maximum Score}) \times 100$

9. Overall Screening Score Examples

Strong Candidate

Total Score: 278

Maximum Score: 300

Overall Score: 92.67

Questions: 3

Average Candidate

Total Score: 210

Maximum Score: 300

Overall Score: 70.0

Questions: 3

Weak Candidate

Total Score: 120

Maximum Score: 300

Overall Score: 40.0

Questions: 3

Empty-score and invalid-input validation were also successfully tested.

Test 4 - Empty Scores: PASSED

Test 5 - Invalid Input: PASSED

STEP 4 TEST: PASSED

10. Step 5 – Score Interpretation

The fifth step converts the numerical screening score into an understandable candidate performance category.

The implemented categories are:

Score Range Category Label

85–100	Excellent	Excellent
70–84	Good	Good
50–69	Average	Average
0–49	Weak	Weak

11. Score Interpretation Descriptions

Excellent

Description:

Candidate demonstrates a very strong screening performance.

Good

Description:

Candidate demonstrates a strong screening performance.

Average

Description:

Candidate demonstrates an acceptable but improvable screening performance.

Weak

Description:

Candidate demonstrates a weak screening performance.

12. Score Interpretation Testing

The following scenarios were successfully tested:

Score: 92

Category: excellent

Label: Excellent

Score: 78

Category: good

Label: Good

Score: 65

Category: average

Label: Average

Score: 40

Category: weak

Label: Weak

Boundary values were also tested:

85 → Excellent

70 → Good

50 → Average

Invalid score validation was successful.

Test 8 - Invalid Score: PASSED

STEP 5 TEST: PASSED

13. Step 6 – Final Screening Scoring Integration

The final step integrates all previous scoring components into a single screening evaluation workflow.

The final scoring engine performs the following operations:

Candidate Responses



Per-Question Scoring



Question Score Aggregation



Overall Score Calculation



Score Interpretation



Final Screening Result

The final result contains:

- Question scores
- Total score
- Maximum score
- Overall score
- Number of questions evaluated
- Category
- Label
- Description

14. Final Screening Result Structure

The final screening result follows this structure:

```
{  
  "question_scores": [...],  
  "total_score": 280,  
  "maximum_score": 300,  
  "overall_score": 93.33,  
  "questions_evaluated": 3,  
  "category": "excellent",  
  "label": "Excellent",  
  "description": "Candidate demonstrates a very strong screening performance."  
}
```

}

This provides a structured output that can be consumed by downstream recruitment systems.

15. Final Integration Testing

Three complete candidate scenarios were tested.

Strong Candidate

Question scores:

Question 1 = 93

Question 2 = 92

Question 3 = 95

Final result:

Total Score: 280

Maximum Score: 300

Overall Score: 93.33

Category: Excellent

Label: Excellent

Questions: 3

Average Candidate

Question scores:

Question 1 = 69

Question 2 = 68

Question 3 = 70

Final result:

Total Score: 207

Maximum Score: 300

Overall Score: 69.0

Category: Average

Label: Average

Questions: 3

Weak Candidate

Question scores:

Question 1 = 34

Question 2 = 32

Question 3 = 32

Final result:

Total Score: 98

Maximum Score: 300

Overall Score: 32.67

Category: Weak

Label: Weak

Questions: 3

The weak candidate total is calculated as:

$$34 + 32 + 32 = 98$$